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COLLEGE OF ENGINEERING, KALIKIRI

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Final Year Project
on

COMPARISON OF BER OF DIFFERENT CHANNEL CODING TECHNIQUES FOR ADVANCED 5G & 6G NETWORKS

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LIST OF CONTENTS

- Abstract
- Introduction
- Problem Statement and Objectives
- Existing Method
- Results of Existing Method
- Disadvantages of Existing Method
- Proposed Method
- Hardware and Software requirements
- Block Diagram
- Channel Coding Techniques
- Implementation and Working Process
- Results Overview
- Comparison of Existing system and proposed system
- Advantages and application of proposed method
- Conclusion

ABSTRACT

In wireless communication systems, data transmission is affected by **noise, interference, and fading**, which increases the Bit Error Rate (BER). Channel coding is an effective technique used to achieve reliable **wireless communication** with a low error rate. In different generations of wireless communication systems, various channel coding techniques such as, **convolutional codes**, **turbo codes**, have been used. However, advanced **5G and 6G communication systems** require channel coding methods that provide lower **Bit Error Rate (BER)** with reduced **computational complexity**.
This project focuses on comparing the BER performance of different coding techniques such as **Convolutional codes, and Low-Density Patity-Check (LDPC) codes.**
Simulation results show that **LDPC codes** provide better **BER performance** with lower complexity compared to other coding techniques and making it suitable for advanced 5G and 6G networks. For further analysis, **turbo codes** and **polar codes** are also implemented, and the performance of all four coding techniques is compared based on BER.

INTRODUCTION

Wireless communication systems are widely used in modern applications such as **mobile communication, IoT, and satellite systems.** However, during transmission, signals are affected by noise and interference, leading to errors in received data. So, the Channel coding plays a key role in improving data reliability by adding redundancy to the transmitted data.With the rapid evolution of mobile communication from 1G to 5G, there is a strong demand for **high data rates, better Quality of Service (QoS), and efficient use of available bandwidth.** In next-generation 6G systems, achieving high reliability and low latency is very important.

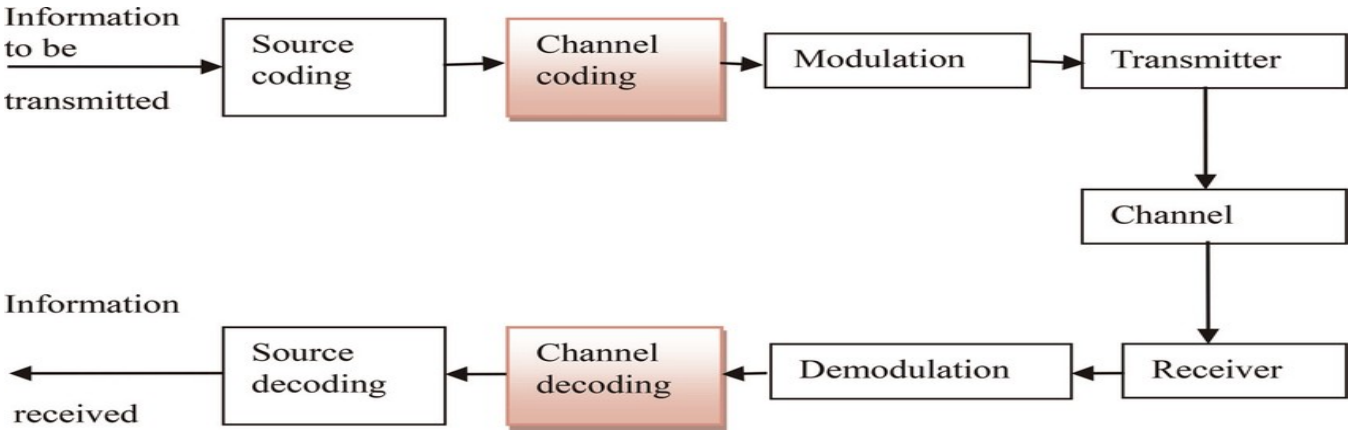


Fig: Block Diagram of Digital Communication System

Problem Statement

In wireless communication systems, signal transmission is affected by noise, interference, and channel fading, **which increases the Bit Error Rate (BER).**

As communication systems evolve towards 6G, there is a demand for:

- Ultra-reliable communication
- High data rates
- Low latency

Different channel coding techniques provide different levels of **performance and complexity.**

Therefore, it is necessary to analyze and compare these techniques to identify the most efficient method that **minimizes BER** while maintaining system efficiency.

Objectives

- To design a communication system model
- To study different channel coding techniques
- To simulate the system using MATLAB
- To analyze BER performance of different techniques
- To identify the best coding technique for advanced 5G 6G networks

Existing Method:

The existing system uses a **MIMO-OFDM communication framework**, where **Orthogonal Frequency Division Multiplexing (OFDM)** divides high data rate signals into multiple low-rate subcarriers to improve **spectral efficiency** and reduce **inter-symbol interference**. To further enhance performance, **Multiple-Input Multiple-Output (MIMO)** technology is used.

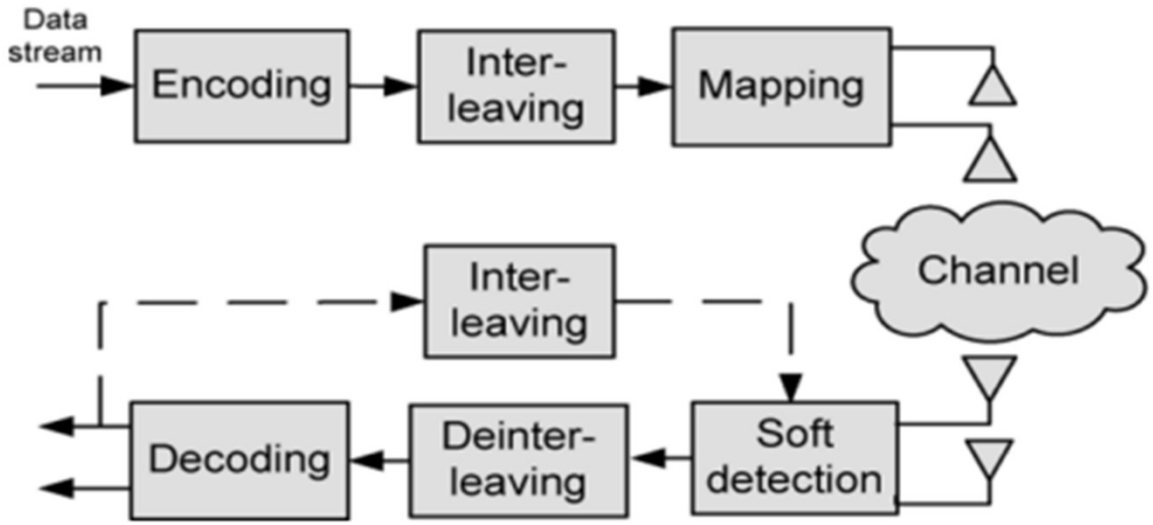


Fig: Block Diagram of Existing Method

The system processes data through **encoding, interleaving, modulation**, transmission over a **noisy channel**, and **decoding** at the receiver. Although this approach improves performance and reduces errors, it has limitations in handling noise efficiently and achieving very low **Bit Error Rate (BER)** required for advanced **5G and 6G communication systems**.

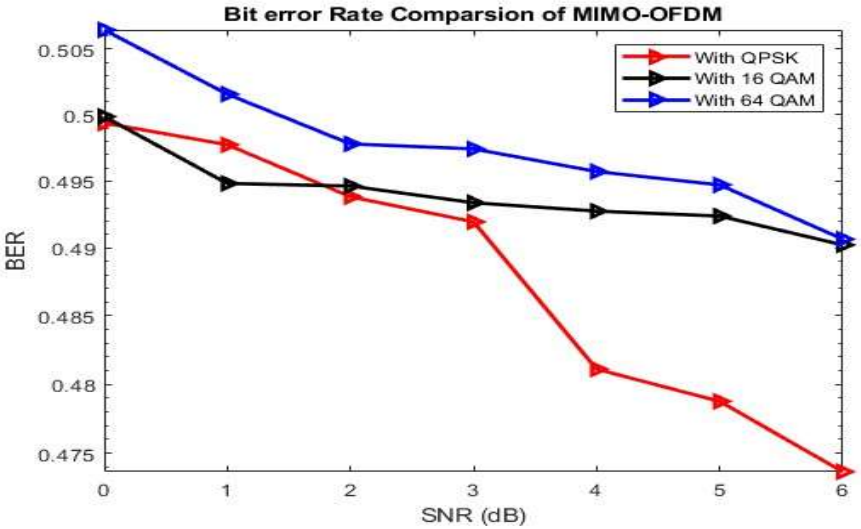
The relationship between data rate and symbol duration is given by:

$$\mathbf{R = 1 / Tsym}$$

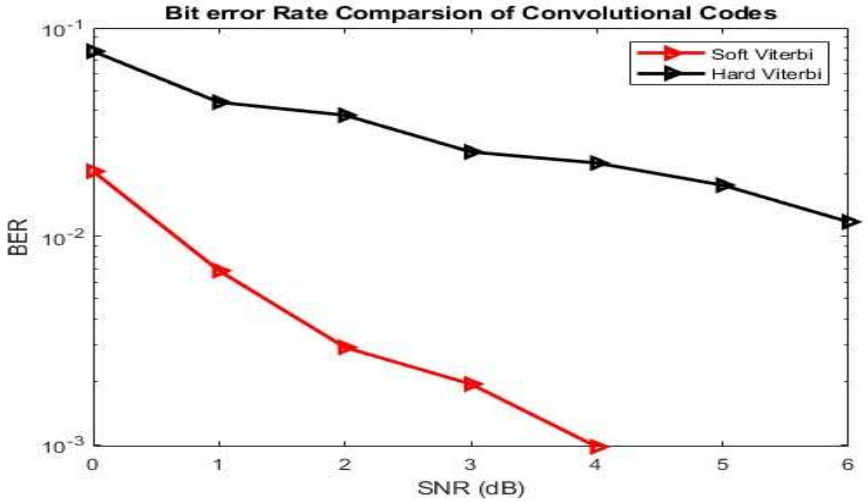
Where:

- R = Data rate (symbols per second)
- Tsym = Symbol duration (time taken to transmit one symbol)

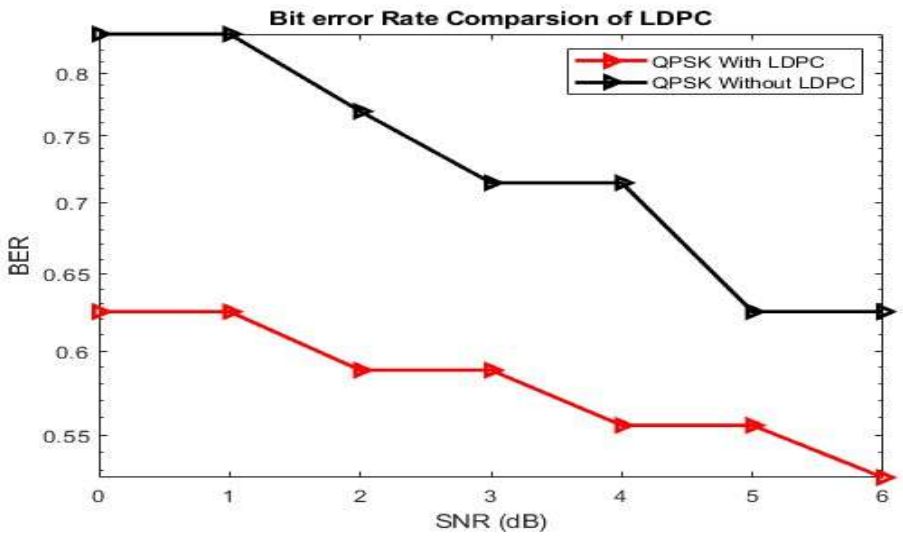
RESULTS OF EXISTING SYSTEM



The graph shows the BER performance of the MIMO-OFDM system using QPSK, 16-QAM, and 64-QAM. As SNR increases, BER decreases for all modulation techniques. QPSK provides better performance with lower BER, while 64-QAM has higher BER due to noise sensitivity. This shows a trade-off between data rate and error performance.



The graph shows the BER performance of convolutional codes using hard and soft Viterbi decoding. Soft Viterbi decoding provides lower BER compared to hard decoding. As SNR increases, BER decreases in both cases. Soft decoding performs better as it uses probability information instead of binary decisions.



The graph shows the BER performance of the system with and without LDPC coding. It is observed that LDPC coding provides lower BER compared to the uncoded system. As SNR increases, BER decreases in both cases. LDPC improves error correction and enhances system reliability, especially at low SNR conditions.

From the above results, it is observed that system performance improves as SNR increases. Advanced techniques like LDPC and soft Viterbi decoding reduce BER compared to basic methods. However, the existing system still cannot achieve very low BER under all conditions, indicating the need for more efficient coding techniques.

DISADVANTAGES OF EXISTING SYSTEM

- High computational complexity
- Poor performance at low SNR (high BER)
- Higher-order modulation (64-QAM) is noise-sensitive
- Hard Viterbi decoding gives lower performance
- LDPC does not achieve very low BER in all cases
- Increased system complexity due to MIMO
- Requires accurate synchronization
- Does not reach Shannon limit performance

Proposed Method:

The proposed system integrates MIMO-OFDM technology with advanced channel coding techniques such as **Convolutional, Turbo, LDPC, and Polar codes** to improve communication performance. It is designed to achieve **low Bit Error Rate (BER)**, **high spectral efficiency**, and **low latency**, which are essential for **5G and 6G systems**. The system starts with random binary data, which is encoded, interleaved, and modulated using **QPSK/QAM**, and then transmitted through a MIMO-OFDM system over an **AWGN channel**. At the receiver, **soft detection**, **deinterleaving**, and **decoding** are performed to recover the original data. Finally, **BER** is calculated and compared for different coding techniques to determine the most efficient method.

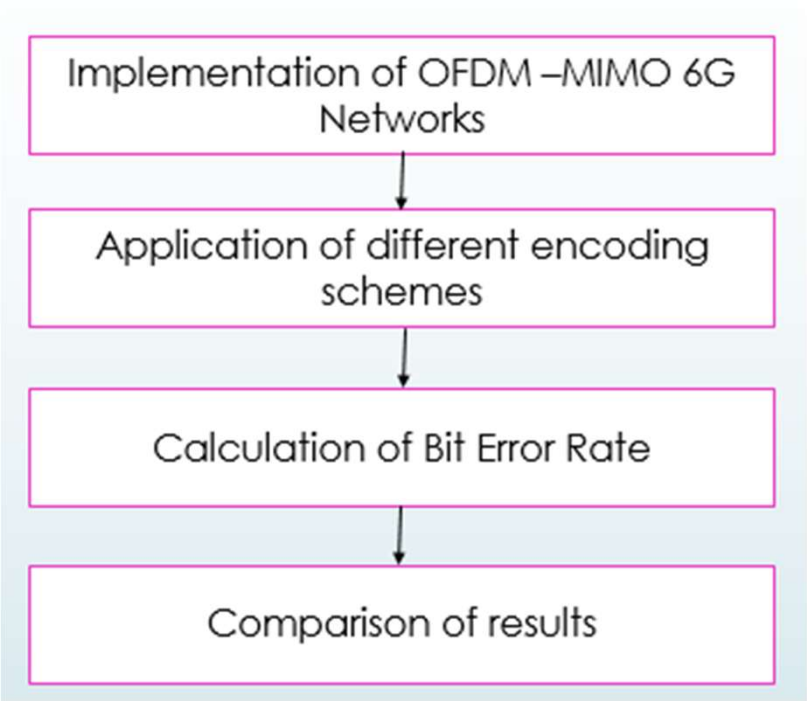


Figure : Flow Chart of Proposed Method

Hardware & Software Requirements

Software: Matlab R2022b

Hardware:

Operating Systems:

• Windows 10

RAM:

Minimum: 4 GB

Recommended: 8 GB

Block Diagram:

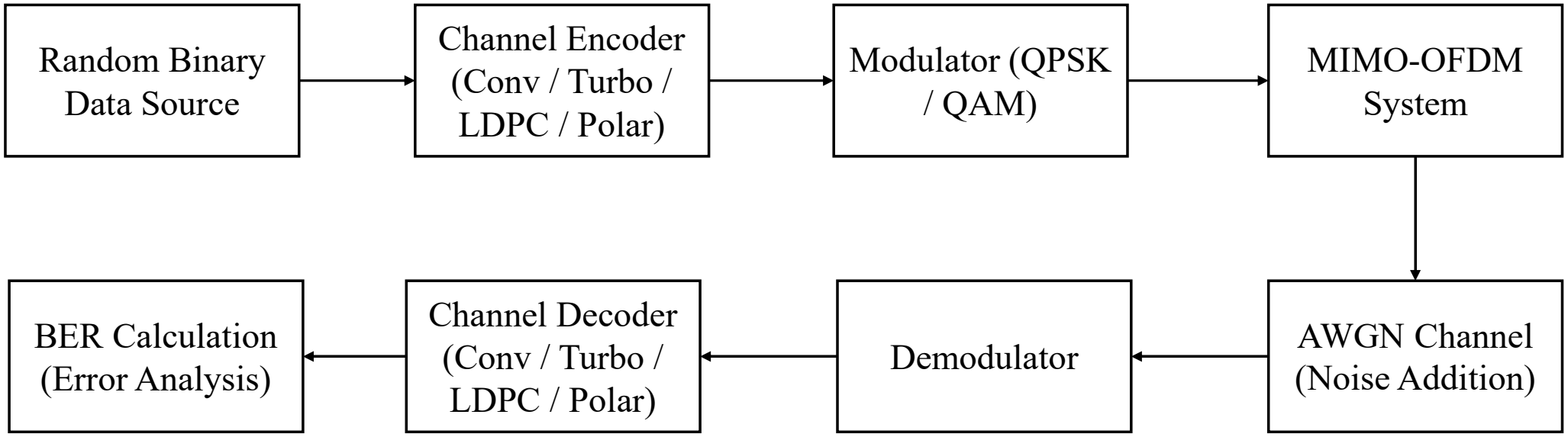


Fig : Block Diagram of Proposed System

METHODOLOGY

The proposed system is implemented using MATLAB simulation to evaluate the performance of different channel coding techniques.

The methodology consists of the following steps:

1. Generate random binary input data
2. Apply channel coding techniques:
    - Convolutional Coding
    - LDPC Coding
    - Turbo Coding
    - Polar Coding
3. Modulate the encoded data using QPSK / QAM
4. Transmit the signal through a noisy channel (AWGN)
5. Demodulate the received signal
6. Decode the signal using respective decoding algorithms
7. Calculate Bit Error Rate (BER)
8. Plot BER vs SNR graph for comparison

Channel Coding Techniques

1. CONVOLUTIONAL CODES

Convolutional codes are widely used error-correcting codes in communication systems, where the output depends on both present and past input bits. They use shift registers and generator polynomials to add redundancy, enabling error detection and correction.

Formula:

$$R = k/n$$

2. LDPC CODES

LDPC (Low-Density Parity-Check) codes are advanced error-correcting codes that use a sparse parity-check matrix for efficient encoding and decoding. They provide very low BER and performance close to the Shannon limit, making them suitable for 5G and modern communication systems.

Formula:

$$H * c^T = 0$$

3. TURBO CODES

Turbo codes are powerful error-correcting codes that use two convolutional encoders and an interleaver to improve performance. The interleaver rearranges input bits, reducing burst errors. Turbo codes achieve very low BER and performance close to the Shannon limit, but they have higher latency and complexity.

Formula:

$$\mathbf{LLR = \log (P(y|x=1) / P(y|x=0))}$$

4. POLAR CODES

Polar codes provide low latency, high spectral efficiency, and excellent performance, making them suitable for next-generation (5G & 6G) systems, though they involve higher complexity. The channels are divided into reliable and unreliable channels, where reliable channels carry information bits and others are assigned frozen bits to improve error correction.

Formula:

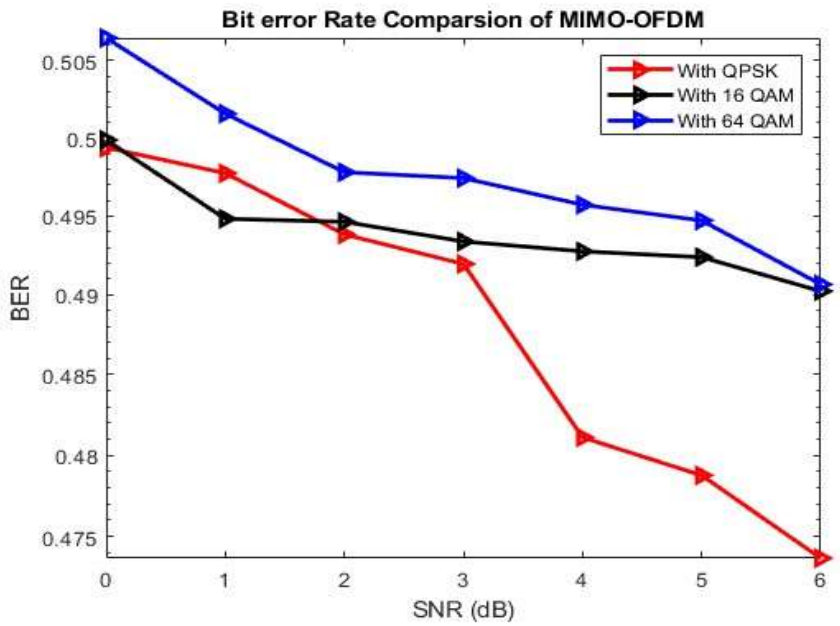
$$\mathbf{x = u * G_N}$$

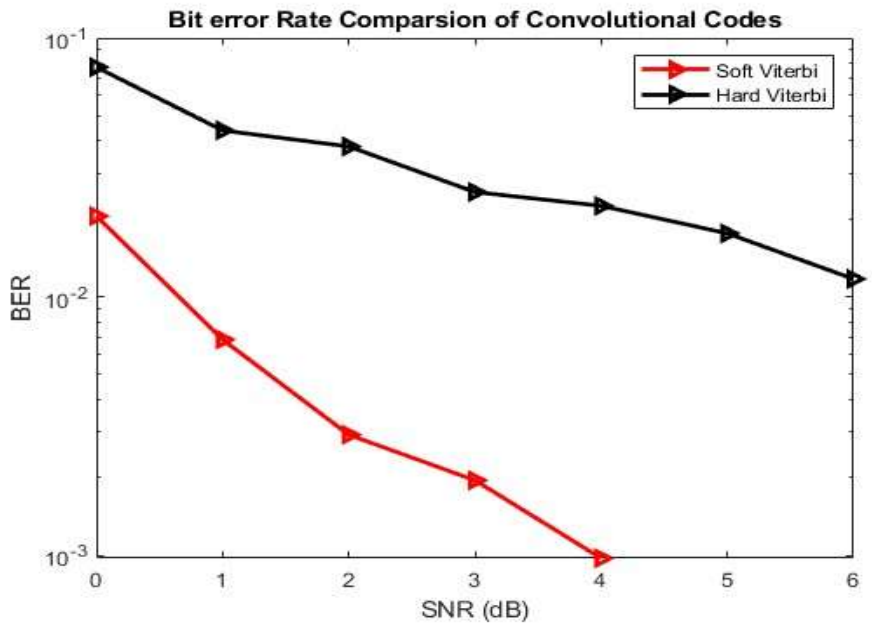
IMPLEMENTATION & WORKING PROCESS

1. The system is implemented in **MATLAB** using a MIMO-OFDM communication model
2. **Random binary data** is generated and encoded using **Convolutional, Turbo, LDPC, and Polar codes** to add redundancy for error correction
3. Encoded data is **interleaved** to reduce burst errors and then modulated using **QPSK/QAM**, converting bits into symbols
4. In **OFDM processing**, **IFFT** converts signals to time domain and a **cyclic prefix** is added to reduce inter-symbol interference
5. The signal is transmitted using **MIMO antennas**, improving data rate, capacity, and reliability, and passes through an **AWGN channel**
6. At the receiver, cyclic prefix removal, **FFT, and signal detection (ZF/MMSE)** are performed to recover the transmitted signal
7. The signal is then **demodulated, deinterleaved, and decoded** using respective decoding techniques to retrieve original data
8. Finally, **BER is calculated** to evaluate and compare the performance of different coding techniques under varying **SNR conditions**

BER = Number of Error Bits / Total Number of Bits

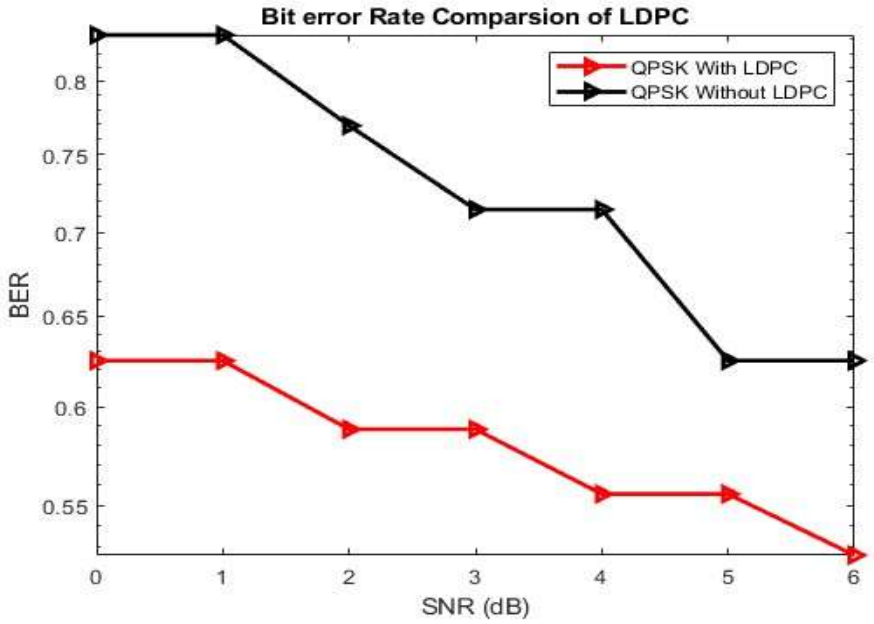
Results Overview

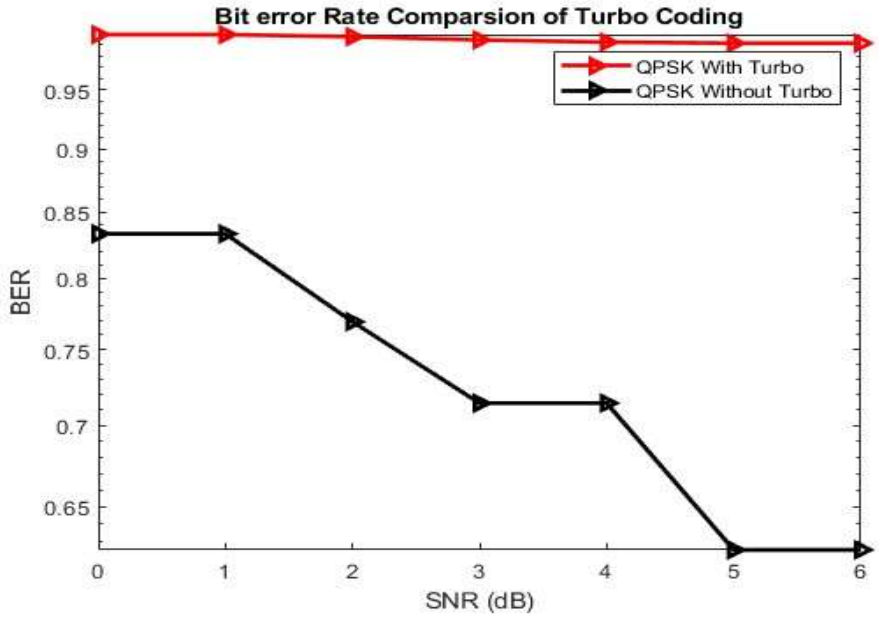




- BER decreases as SNR increases. QPSK shows lowest BER, while 64-QAM has higher BER due to noise sensitivity. Lower-order modulation is more reliable.

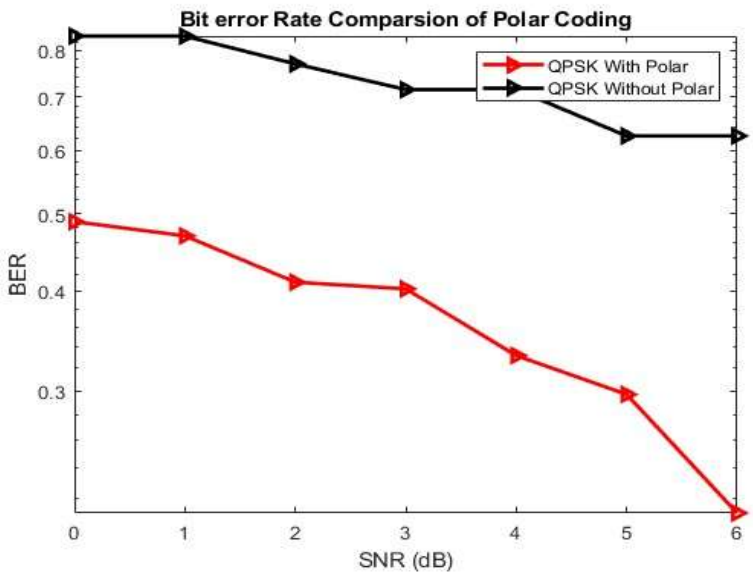
- Soft Viterbi decoding gives lower BER than hard decoding. BER decreases with SNR. Soft decoding improves performance using probability information.





- LDPC significantly reduces BER compared to without coding. Performance improves as SNR increases, showing strong error correction capability.

- Turbo coding improves BER gradually with SNR. It provides good error correction but has higher complexity and latency due to iterative decoding.



- Polar coding shows the best performance with lowest BER. BER decreases rapidly with SNR, making it suitable for 5G and 6G systems.

From the proposed system graphs, it is observed that Advanced channel coding techniques significantly enhance performance. **Soft Viterbi decoding**, **LDPC**, and **Turbo codes** improve error correction, while **Polar coding shows the best performance** with the lowest BER, low latency, and high efficiency, making it most suitable for **advanced 5G and 6G communication systems**.

COMPARISON OF EXISTING AND PROPOSED SYSTEM

Feature	Existing System	Proposed System
Coding Techniques	Convolutional, LDPC	Turbo, Polar
BER Performance	Moderate	Very Low
Latency	Higher	Lower
Spectral Efficiency	Limited	High
Performance at Low SNR	Less efficient	More efficient
Complexity	Moderate	Higher but optimized
Reliability	Moderate	High
Suitability for 5G/6G	Not Suitable	Highly Suitable

SUITABILITY OF CODING TECHNIQUES FOR COMMUNICATION SYSTEMS

Coding Technique	Suitable Communication Systems
Convolutional Coding	2G, 3G systems, basic wireless communication, satellite communication
LDPC Coding	4G LITE and Wi-Fi systems
Turbo Coding	5G Systems, deep space communication, mobile networks
Polar Coding	Advanced 5G control channels, advanced wireless systems, future 6G networks

Advantages and Applications of Proposed Method

Advantages

- Reduced Bit Error Rate (BER)
- Improved reliability and error correction
- High spectral efficiency using MIMO-OFDM
- Better performance at low SNR
- Low latency with Polar coding
- Supports high data rate communication

Applications

- 5G and 6G communication systems
- Mobile and wireless networks
- Satellite communication
- IoT and Wireless Sensor Networks
- High-speed data transmission
- Military and secure communication

CONCLUSION

This project analyzes different channel coding techniques such as Convolutional, Turbo, LDPC, and Polar codes using a MIMO-OFDM system. The results show that the **proposed system**, which includes advanced coding techniques like **Turbo and Polar codes**, achieves improved performance in terms of reduced Bit Error Rate (BER), better reliability, and higher spectral efficiency compared to the existing system. Among all techniques, **Polar coding** provides the best performance due to its low latency and efficiency, making it suitable for advanced **5G and 6G communication systems**. Overall, the proposed system enhances communication performance and offers a reliable solution for **next-generation wireless networks**.

The study also demonstrates the importance of **advanced error correction techniques** in improving system efficiency. It highlights the need for **low latency and high data rate communication** in modern wireless systems.

References

[1] S. Bates et al., “Termination sequence generation circuits for LDPC convolutional codes,” IEEE Trans., 2005.

[2] S. Bates et al., “Decoders for LDPC convolutional codes,” NASA VLSI Symposium, 2005.

[3] S. M. Motaghiannezam et al., “High-speed PAM4 transmission,” OFC Conference, 2016.

[4] Additional MATLAB and communication system resources

THANK YOU